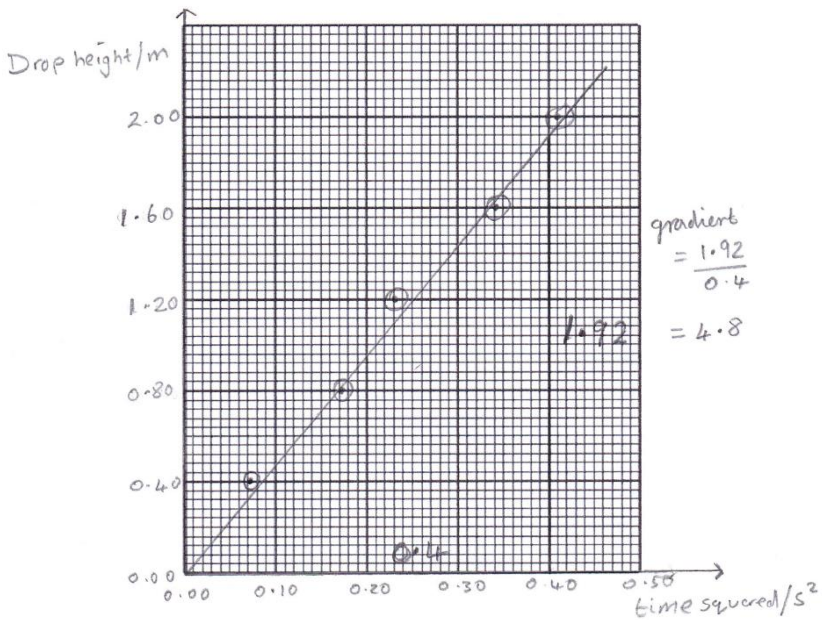
<table>
<tr><th colspan="3" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="4">Marks available</th><th></th><th></th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>2</td><td>(a)</td><td>(i)</td><td>Magnitude of vertical force = 2.0×10^{-4} (N) (1) [Direction not required here]
Application of Pythagoras and correct overall magnitude determined i.e $R^2 = (2.0 \times 10^{-4})^2 + (5.0 \times 10^{-4})^2$
$R = 5.4 \times 10^{-4}$ (N)[accept 5.38 or 5.39] (1) No ecf
$\theta = 21.8°$ [tolerance of rounding errors] below horizontal [allow 112°, 22° South of E…] stated or clearly shown in diagram [accept 68.2° to the vertical stated or shown] ecf on R (1)</td><td></td><td>3</td><td></td><td>3</td><td>3</td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td>Air resistance and force due to gravity [or weight] are equal [or air resistance = 6.0×10^{-4} N] hence no resultant force [accept forces balanced /cancel / no acceleration]</td><td>1</td><td></td><td></td><td>1</td><td></td><td></td></tr>
<tr><td></td><td>(b)</td><td>(i)</td><td>Subtract …0.05 [from readings (of time)] / ….the time delay / …it</td><td></td><td></td><td>1</td><td>1</td><td></td><td>1</td></tr>
<tr><td></td><td></td><td>(ii)</td><td><table>
<tr><td>Drop height h, (m)</td><td>0.40</td><td>0.80</td><td>1.20</td><td>1.60</td><td>2.00</td></tr>
<tr><td>Corrected time t, (s)</td><td>0.27</td><td>0.41</td><td>0.48</td><td>0.58</td><td>0.64</td></tr>
<tr><td>Corrected time squared t^2, (s^2)</td><td>0.07(3)</td><td>0.17</td><td>0.23</td><td>0.34</td><td>0.41</td></tr>
</table>
All values of t^2 calculated correctly (1) [award this mark, even if sig figs incorrect]
To 2 sig. fig. [Allow 1 sf on first answer] [see table] (1)</td><td></td><td>2</td><td></td><td>2</td><td>2</td><td>2</td></tr>
<tr><td></td><td></td><td>(iii)</td><td>$x = ut + ½at^2$ identified (1)
Explanation that: [$x = h$], $u = 0$ and $a = g$ [or by implication] (1)
No algebra required</td><td>1</td><td>1</td><td></td><td>2</td><td></td><td></td></tr>
</table>

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iv)	Suitable scale and both axes labelled correctly with appropriate units: [drop] height [h]/ m and time squared [t^2]/ s^2 [or $(t$ / s$)^2$] (1) Allow ecf from table All 5 points plotted correctly ± ½ small square division (2) If 4 points plotted correctly ± ½ small square division (1) If 3 or less points plotted correctly ± ½ small square division (0) Appropriate line of best fit [through origin] (1) **ecf** 		4		4	4	4
		(v)	Suitable triangle shown on graph [or two points on line implied in calculation] with $\Delta h \geq 1.0$ m [or two appropriate points shown on line] See above (1) Gradient calculated correctly [Accept 4.6 to 5.0] (1) $g = 2 \times$ gradient [Allow **ecf**] (1) 2nd and 3rd mark can be awarded even if first mark withheld. [Allow final mark for correct answers using data points rather than gradient]			3	3	2	3

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)		Straight line / $h \propto t^2$ / linear graph (1) Through [or close to] origin (1) g close to accepted value / 9.81 or low degree of scatter / points close to line of best fit [accept relevant comment based upon candidate's graph] (1)			3	3		3
			Question 2 total	**2**	**10**	**7**	**19**	**11**	**13**